

Table A: Task A — Pairwise Classification (AUCROC)

Model	Method	Repr	Best Layer	AUCROC
LaTa	Baseline	hidden	12	0.9292
LaTa	Baseline	ff1	1	0.8160
LaTa	SIF only	hidden	1	0.9265
LaTa	SIF only	ff1	1	0.9394
LaTa	SIF+ABTT (opt D)	hidden	6	0.9669
LaTa	SIF+ABTT (opt D)	ff1	12	0.9671
LaTa	Whitening	hidden	4	0.5406
LaTa	Whitening	ff1	4	0.5598
PhilTa	Baseline	hidden	12	0.9132
PhilTa	Baseline	ff1	11	0.7881
PhilTa	SIF only	hidden	1	0.9459
PhilTa	SIF only	ff1	7	0.9378
PhilTa	SIF+ABTT (opt D)	hidden	12	0.9668
PhilTa	SIF+ABTT (opt D)	ff1	12	0.9550
PhilTa	Whitening	hidden	6	0.5247
PhilTa	Whitening	ff1	5	0.5747
LaBSE	Baseline	hidden	12	0.9430
LaBSE	SIF only	hidden	12	0.9420
LaBSE	SIF+ABTT (opt D)	hidden	12	0.9782
LaBSE	Whitening	hidden	2	0.5560
Qwen3-0.6B	Baseline	hidden	28	0.9519
Qwen3-0.6B	Baseline	ffn_int	20	0.9464
Qwen3-0.6B	SIF only	hidden	28	0.9308
Qwen3-0.6B	SIF only	ffn_int	20	0.9354
Qwen3-0.6B	SIF+ABTT (opt D)	hidden	3	0.9688
Qwen3-0.6B	SIF+ABTT (opt D)	ffn_int	2	0.9669
Qwen3-0.6B	Whitening	hidden	3	0.5780
Qwen3-0.6B	Whitening	ffn_int	2	0.5750
KaLM-mini	Baseline	hidden	23	0.9584
KaLM-mini	Baseline	ffn_int	19	0.9436
KaLM-mini	SIF only	hidden	23	0.9512
KaLM-mini	SIF only	ffn_int	19	0.9408
KaLM-mini	SIF+ABTT (opt D)	hidden	3	0.9573
KaLM-mini	SIF+ABTT (opt D)	ffn_int	2	0.9653
KaLM-mini	Whitening	hidden	18	0.5779
KaLM-mini	Whitening	ffn_int	6	0.5342